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BASED ON REAL-WORLD CYBERSECURITY INCIDENTS.**

**AI-DRIVEN ATTACKS IN MODERN POLITICS: THE 2024 U.S. ELECTION DEEPPAKE CASE
STUDY.**

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PART A

1.0 REAL-WORLD CYBERSECURITY INCIDENT

During the 2024 Presidential election in the USA, experts, researchers and government officials in cybersecurity continue to raise alarms regarding misinformation and AI generated manipulation via Social Media Platforms. Campaigns are no longer limited to traditional mediums of communication, such as TV ads and speeches - Political Campaigns have now transitioned to using all of the following Social Media Platforms for political purposes — Twitter (X), Instagram, Facebook, TikTok, WhatsApp, YouTube etc. Daily, millions of users post and share political beliefs, political advertisements, video content and news to/through these social media platforms, thus also allowing potentially bad actors to distribute misinformation rapidly to huge audiences. Researchers also indicate that bots, fake accounts and algorithmic recommendation systems all support political related content being distributed across the world/in the US today. [10]

One of the major issues during the election is the rise of “Deep Fake” content and artificially created Fake Political Content. Some Deep Fake videos, fake audio recordings, manipulated speeches and images generated through AI will intend to mislead, sway or influence voter opinion/electorate. A recent example of this type of content is fake robocalls imitating President Joe Biden’s voice were spread online to discourage voting, while AI-generated political propaganda and manipulated videos circulated widely on social media. Experts warn, with the evolution of AI technology, that creating, distributing and propagating misinformation on a political subject is becoming much easier through the use of Artificial Intelligence due to the production of extremely realistic fake content being created instantaneously through the use of modern AI systems. [10]

At the same time, researchers observed that social media bots and trolls played a major role in spreading political misinformation. The automated or fake accounts use emotional and aggressive messages related to politics to engage with others and influence conversations taking place within the digital marketplace. Furthermore, many of these bots can be imitated as authentic people and thus are challenging to detect. Studies have also shown that the targeted advertisement of political campaigns occurs extensively in election years. Online political advertisements can utilize large quantities of publicly available data; therefore, online campaigns and actors are capable of targeting specific groups of voters with tailored messages in order to enhance political polarization and manipulation.[10]

The incident raised serious concerns about election integrity, public trust, and cybersecurity. Many experts feared that voters would struggle to distinguish between real and fake information online. Governments, researchers, and social media companies responded by discussing stronger AI regulations, deepfake detection systems, fact-checking mechanisms, and laws requiring disclosure of AI-generated political content.[10]

2.0 ATTACK LIFECYCLES

i. Initial Access

The attackers gained their foothold mainly through social media platforms and AI-generated content. In order to accomplish this, the attacker can create false identities through fake accounts, bots, deepfake videos, manipulated audio (voice or facial recognition) and create political ads through AI technology, all of which can be found on Twitter/X, Facebook, TikTok, and WhatsApp, which are now the primary channels used by attackers to push false information, as well as influence the opinion of the general public.

ii. Privilege Escalation

The attacker was able to increase their influence by using automated bots, fake viral posts and algorithmic manipulation. The creation of convincing AI-authored content that appears to be real allows the attackers/creator to deceive users more easily, while the engagement generated through likes, shares and comments further encourages social media algorithmic promotion, providing the attacker with increased privilege in online political discussions.

iii. Lateral Movement

The misinformation spread rapidly across multiple social media platforms and online communities. Use of bots and fake accounts to continually repost and amplify political propaganda allows for the dissemination of the attack across social media platforms. Misinformation can rapidly go from Twitter/X to Facebook, TikTok, WhatsApp groups, YouTube and online communities. This cross-platform sharing of misinformation has the ability to enhance political polarization and expand the attack's influence.

iv. Data Mining

Attackers no longer just take regular private data; they mostly take user behavior, political views, online relationships and demographics. This information is then used to target political advertising and manipulate voters. In addition, Artificial Intelligence (AI) and big data analytics are used to help analyze voters and deliver customized political messages to influence how people feel and how they vote during elections.

3.0 ANALYSIS OF SUCCESSFUL ATTACKS RELATED TO MODERN ISSUES

Most of the success of the misinformation and AI-driven political manipulation campaign can be linked to several modern cybersecurity and digital communication challenges. The attack did not rely on traditional system exploitation but instead combined artificial intelligence, social media algorithms, interconnected digital platforms, and human psychological factors to influence public perception and voter behaviour. These factors made the attack highly effective and difficult to contain. [8]

3.1 AI-Driven Content Generation as a Modern AI Threat

The attack was successful because modern generative AI tools can create highly realistic deepfake videos, cloned voices, and convincing political messages at a very low cost. In the 2024 U.S. Election case, attackers used AI-generated robocalls that imitated President Joe Biden's voice to discourage citizens from voting.

Unlike traditional misinformation campaigns that require significant resources and production time, AI enables attackers to produce large amounts of believable content within minutes. This scalability allows misinformation to spread rapidly before fact-checkers or authorities can respond. As a result, AI significantly lowers the barrier for conducting large-scale influence operations.

3.2 Algorithm Amplification in the Social Media Ecosystem

Social media platforms such as X (Twitter), Facebook, TikTok, and YouTube contributed to the success of the attack through algorithm-driven content promotion. These platforms prioritize content that generates high engagement, including likes, comments, shares, and reposts.

In the election incident, politically charged and emotionally provocative content received greater visibility because users interacted with it more frequently. Attackers further amplified their reach using bots and fake accounts that artificially increased engagement metrics. Consequently, misinformation was promoted

by platform algorithms and reached millions of users without requiring significant advertising expenditure.

3.3 Cross-Platform for Digital Environment creating modern connectivity issue

Modern digital environments are highly interconnected, allowing information to move rapidly between multiple platforms. Content shared on one platform can quickly be reposted or redistributed across others.

In the election case, misinformation initially appearing on X or Facebook could later spread through WhatsApp groups, TikTok videos, YouTube clips, and online communities. This cross-platform dissemination made containment extremely difficult because removing content from one platform did not prevent it from continuing to spread elsewhere. The interconnected nature of modern communication systems therefore increased both the speed and scale of the attack.

3.4 Human trust and cognitive bias

Human psychology was one of the most important factors contributing to the attack's success. Many users tend to trust information shared by friends, community groups, or accounts that appear legitimate.

The Biden voice-cloning incident demonstrated how people are more likely to believe information when it appears to come from a familiar or trusted source. In addition, confirmation bias caused users to accept information that aligned with their existing political beliefs without verifying its authenticity. Attackers exploited these psychological tendencies to increase the credibility and effectiveness of their misinformation campaigns.

4.0 VULNERABILITY IDENTIFICATION

The success of the attack was enabled by a combination of technical, human, and organizational vulnerabilities. These weaknesses created opportunities for attackers to distribute misinformation, manipulate public opinion, and undermine trust in election processes.

4.1 Technical Vulnerability

From a technical point of view, social media platforms do not yet have strong real-time systems to detect AI-generated content like deepfakes, fake audio, or edited text. Even bot detection systems can be tricked by advanced AI that behaves like real humans. In addition, there is no global system that can confirm if a photo, video, or message is truly authentic; and because of this, fake content can spread widely before it is

detected and removed. For example, the AI-generated robocall that imitated President Joe Biden's voice successfully reached voters because existing systems could not immediately verify that the audio was artificially generated. [7]

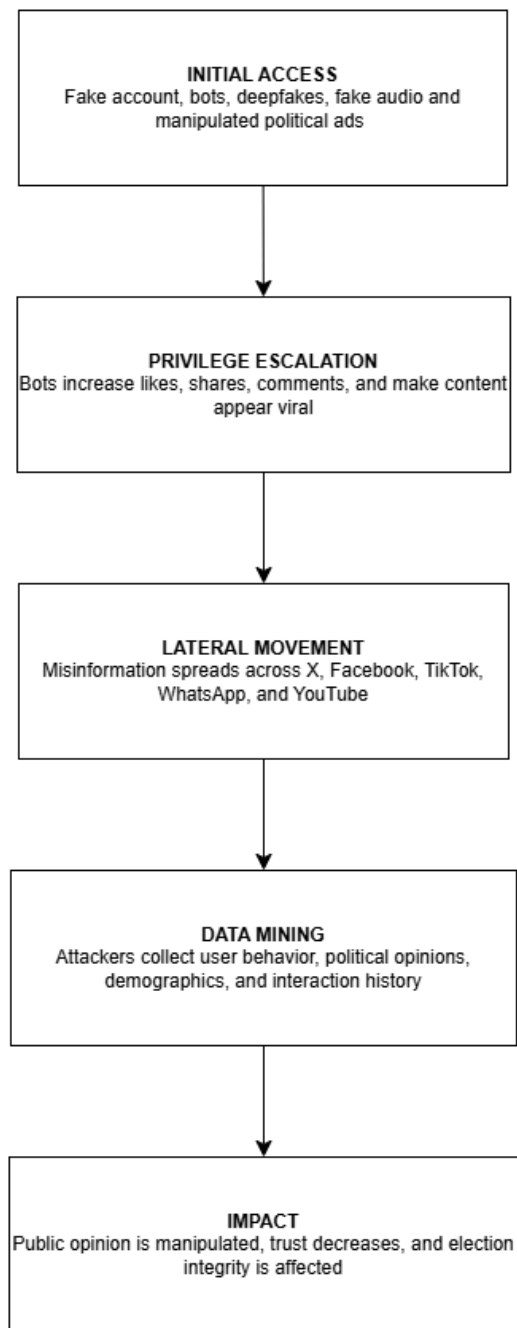
4.2 Human Vulnerability

From a human perspective, users are particularly vulnerable because many lack the skills required to identify fake news, deepfakes, and manipulated content. People often react to emotional or political posts without verifying their accuracy. Many individuals also tend to believe information that supports their existing views, a behaviour known as confirmation bias. As a result, users may unknowingly spread misinformation to others. For example, some voters believed the AI-generated Biden robocall because the voice sounded authentic and appeared to come from a trusted political figure. [7]

4.3 Organizational Vulnerability

From an organizational point of view, weak rules and poor control make the problem even worse. There are not enough strict global laws for controlling AI-generated political content. Thus, social media companies usually act after misinformation spreads instead of stopping it early. Also, political advertising is not fully transparent, this causes the attackers to target specific groups of voters using hidden data and personalized messages. For example, the lack of mandatory verification and disclosure requirements for AI-generated political content allowed manipulated media to be distributed without immediate identification as synthetic content. [7]

5.0 ATTACK FLOW DIAGRAM



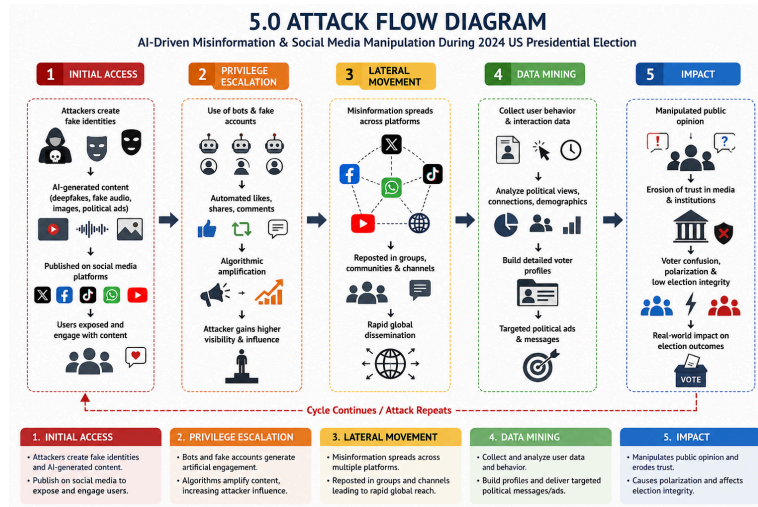


Figure 1: Attack flow diagram

The attack flow diagram explains how AI-driven misinformation attacks were executed during the 2024 U.S. Presidential Election using social media platforms, deepfake technology, automated bots, and data analytics.

5.1 Initial Access

In the first phase, attackers gain initial access to the public by creating fake identities and AI-generated political content.

The attackers use:

Fake social media accounts

Automated bots

- Deepfake videos
- AI-generated audio recordings
- Manipulated images
- Fake political advertisements

These fake contents are distributed through platforms such as:

- X (Twitter)
- Facebook

- TikTok
- WhatsApp
- YouTube

The objective is to expose users to misleading information and encourage them to interact with the content.

Impact:

Users begin to trust, watch, like, comment, and share the manipulated political content without realizing it is fake.

5.2 Privilege Escalation

After gaining attention online, attackers increase their influence using social media algorithms and automated engagement tools.

The attackers:

- Use bots to generate fake engagement
- Increase likes, comments, shares, and reposts
- Make content appear viral and trustworthy

Social media algorithms automatically promote content with high engagement, causing the misinformation to spread even further.

Impact:

The attackers gain stronger visibility, credibility, and influence in political discussions online.

5.3 Lateral Movement

In this stage, the misinformation spreads rapidly across multiple platforms and online communities.

The attackers continuously repost and distribute the fake content through:

- Facebook groups

- WhatsApp communities
- TikTok reposts
- YouTube clips
- X/Twitter threads

Cross-platform sharing allows the misinformation to travel quickly across the internet.

Impact:

- Rapid global dissemination of fake information
- Increased political polarization
- Difficulties in controlling or removing misinformation

This stage is similar to lateral movement in cybersecurity attacks where attackers spread across systems after gaining access.

5.4 Data Mining

Once users interact with the misinformation, attackers collect valuable data from online activities.

The attackers gather:

- User behavior patterns
- Political opinions
- Demographic information
- Online relationships
- Interaction history

AI and big data analytics are then used to analyze voter behavior and create detailed voter profiles.

Impact:

Attackers can send highly targeted political advertisements and personalized propaganda to manipulate voter decisions more effectively.

5.5 Impact

The final phase explains the overall consequences of the attack on society and democratic systems.

The attack causes:

- Manipulation of public opinion
- Spread of confusion and misinformation
- Loss of trust in media and government institutions
- Political polarization among citizens

Final Consequences:

- Election integrity becomes weakened
- Public confidence in democracy decreases
- Voting decisions may be influenced by false information

Overall, the attack demonstrates how AI-generated misinformation and social media manipulation can create serious cybersecurity and political risks in modern elections.

PART B

1.0 ORGANIZATION CONTEXT

1.1 Organization Background

The organization selected in this study is a national election management agency, similar to a national election commission, responsible for managing election-related digital systems, protecting voter information, securing official election communication, and maintaining public trust in the democratic process. Operating in a high-stakes, time-sensitive environment, the agency serves as a important role in ensuring election security, transparency and operational reliability. The modern electoral environment has moved away from mechanical, human-operated workflows, and therefore this agency must rely heavily on the complex ecosystem of cloud infrastructure, digital technologies and online communication platforms to carry out domestic operations and distribute public information. [1]

1.2 Infrastructure and Data Handling

As a national election management agency, the organization implements a large-scale digital architecture to support daily election operations, manage stakeholder communication, and protect sensitive election-related data. The organization manages and safeguards several important categories of sensitive data through these systems, as below: [3] [5]

- Voter and Registration Data: On cloud databases that include personally identifiable information (PII) and the login logs of all registered voters.
- Electoral & Tabulation Records: Some of the most sensitive data sets that include details of real-time and historical election results.
- Official Communications & Media Assets: Digital communication from the official election management website and rapid-response social media networks.
- Internal Operational Data: Private correspondence and administrative information that flows through secure internal networks from election officers to government staff.
- Threat Intelligence Data: Aggregate data from AI-supported media monitoring systems that are used to track & alert on online misinformation and suspicious activities.

1.3 Challenger Environment and Perils to Control

Operating in a multi-linking digital ecosystem that encompasses government agencies, political campaigns and media outlets with third-party cloud vendors and other parties, the agency's attack surface is broadened. Any single connected system vulnerability can potentially destroy infrastructure altogether. Inspired by the rapid developments in Artificial Intelligence (AI) and the standard forms of cyber threats, these threats fall into three broad vectors:

a) Psychological and disinformation risks.

- Deepfake and Media Manipulation: Advanced generative AI tools are used by malicious actors to produce hyper-realistic fake videos, fake speeches and cloned political persona. These are meant to proliferate focused misinformation, stoke voter frustration and undermine public faith in democratic traditions. [6][2]
- Social Engineering and Phishing: With the help of deep learning-driven AI-driven phishing campaigns, human vulnerabilities can be exploited. Employees and election officials can be duped by cloned voices or highly personalized emails to launch bad files, download malware or disclose administrative passwords. [3]

b) Tech & Infrastructure Risks.

- Identity-Based Attacks : Threat actors can impersonate government officials or election officers using stolen credentials, weak authentication bypasses, or AI-generated voice cloning to obtain unauthorized lateral access to internal networks. [3]

- **Data Breaches and Exfiltration:** Things like voter registration databases, election records hosted in the cloud, or confidential internal communications can still be exposed via cloud misconfigurations, zero-day exploits, or credential theft. [4]
- **Distributed Denial of Service (DDoS):** Threat actors can employ coordinated network floods to intentionally bring down official election management sites, public informational repositories and online communication avenues, locking the doors during the election cycle during the voting season and tabulation window.

c) Risks of Affecting Operations and Institutions.

- **Reputation and Democratic Damage:** The effective execution of AI-based cyber or disinformation attacks undermines legitimate perception of the electoral outcome. This can result in broad public confusion, social panic, institutional delegitimization and profound political instability. [6]

1.4 Why the Organization is a High-Value Target

It is a linchpin of national security. State sponsored attack actors and hacktivists and cybercriminals place this organization among classified high value targets (HVT) as a result of the following critical dependencies:

- **Guardian of Key Democratic Information:** As custodian at the core of important, sensitive national infrastructure information, including proprietary voter identity data (PII) and official election tabulation data.
- **Direct Impact on Public Trust and Sovereignty:** Unlike businesses where breaches can lead to monetary damage, the attack inside of this agency attacks the psychological fabric of the nation. Breaking its workflows or altering it even serves to erode the country's confidence in democratic values, thus making it a prime target for foreign adversaries wishing to disrupt a state of geopolitical stability.
- **Social Media is the Weaponized Accelerator:** With an emphasis on its media-fueled public information networks to propagate updates, such that the agency is extremely capable of deploying news via its public communication networks on speed, it becomes incredibly vulnerable to outside threat vectors. AI-generated misinformation or twisted official accounts that have become a standard means of disseminating information can quickly spread across media and social networks, without the time-consuming, human-led, traditional verification teams that human-centered verification teams can rely on for protection that is so crucial for this category.

As this is a special confluence of high stakes data ownership and higher-skill AI threat vectors, legacy boundary defenses are simply not enough as it evolves into an AI frontier. Mitigating these important vulnerabilities, the agency needs to have sophisticated, holistic network defence built upon Zero Trust Architecture (ZTA), advanced Identity and Access Management (IAM) and perpetual Security Information and Event Management (SIEM) controls in place.

2.0 NETWORK SECURITY ARCHITECTURE DIAGRAM

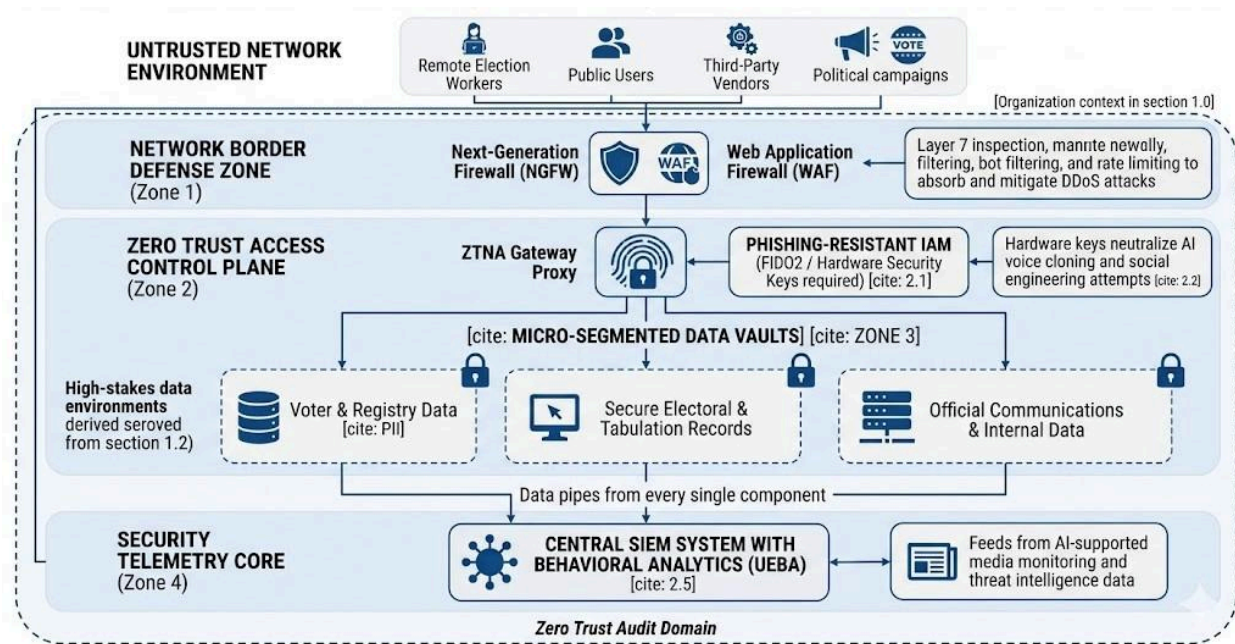


Figure 2 : Network architecture diagram

Diagram Design Features :

1. **Multi-Layered Defense:** Before reaching high-stakes data, traffic passes through a series of security checkpoints after leaving the untrusted public internet (top).
2. **Next-Gen Perimeter:** Examines Layer 7 traffic and filters out automated bots and DDoS threats using Firewalls (NGFW) and Web Application Firewalls (WAF).
3. **Zero Trust Access Control:** Enforces the idea that know trust is never given depend on network location by implement a ZTNA Gateway Proxy.
4. **Identity-Centric Focus:** Incorporates IAM that is resistant to phishing and requires the use of cryptographic hardware security keys. This directly combats the threat posed by deepfake social engineering and sophisticated AI voice cloning.

5. Micro-Segmentation: To prevent lateral movement, critical data environments such as official communications, tabulation records, and voter PII data are rigorously isolated in logical "Vaults" behind extra segmentation walls.
6. Centralized Intelligence: To identify rapid automated anomalies, each component transmits telemetry into a central SIEM system that has been upgraded with User and Entity Behavior Analytics (UEBA).

3.0 RECOMMENDED SECURITY TECHNOLOGIES

To safeguard our public sector government election management agency against advanced generative AI exploits and infrastructure-level threats, four core security technologies have been selected as a layered defense matrix.

3.1 Zero Trust Architecture (ZTA) via ZTNA Gateways

A. Justification

Perimeter defense (also known as "castle-and-moat") is the foundation of traditional network security, which implicitly trusts everyone inside the network boundary. The 2024 election issue showed how contemporary threat actors completely circumvent perimeters by taking advantage of cloud setups and human trust. A legacy network would enable an attacker to migrate laterally and compromise the voter database if they used an AI voice clone or a hyper personalized phishing email to trick a distant election worker or a system administrator. Implicit trust is eliminated by zero trust. It guarantees that the adversary is structurally constrained and cannot access the extremely important Electoral & Tabulation Records, even in the event that an identity or endpoint is compromised by an AI social engineering assault.

B. Function

The agency implements Zero Trust Network Access (ZTNA) using an identity-aware gateway proxy configured on a principle of "Never Trust, Always Verify."

1. 1. Continuous Context Verification: The ZTNA gateway intercepts each request made by an election officer for access to an internal application, such as the voter registration ledger. It dynamically assesses the user's explicit identity, the exact location, the time of the request, and the security posture of the device (e.g., determining whether Endpoint Detection and Response [EDR] is activated).
2. Micro-Segmentation: Separate security zones are created within the internal cloud infrastructure. The gateway creates temporary, encrypted one-to-one connections between the verified user and the particular program they are permitted to utilize.
3. Preventing Lateral Movement: The endpoint cannot see the rest of the network since it is entirely dark. The micro-segmentation barriers stop network discovery and lateral pivoting into the tabulation databases in the event that an attacker takes control of a low-privilege communications laptop.

3.2 Phishing-Resistant Identity & Access Management (IAM) via FIDO2

A. Justification

The Al Jazeera source highlights that generative AI tools can instantly synthesize highly realistic voice clones of trusted authorities for a trivial cost (\$1) [9]. In a high-stakes election environment, an attacker can use a voice clone of a senior director to call an on-duty technical officer and order them to disclose a password or read aloud a text-based SMS/Email One-Time Password (OTP). Traditional Multi-Factor Authentication (MFA) methods like mobile push notifications, SMS codes, or authenticator apps are completely vulnerable to this category of AI-driven social engineering and "MFA fatigue" attacks. Phishing-resistant IAM is chosen because it removes human susceptibility from the authentication loop entirely.

B. Function

The agency mandates enterprise-wide deployment of FIDO2 / WebAuthn standard hardware security across all administrative accounts.

1. Cryptographic Binding: When an employee attempts to log into an agency system, the FIDO2 credential architecture binds the cryptographic login key directly to the specific, legitimate domain name of the election agency.
2. Asymmetric Cryptography: The user must physically touch the hardware key plugged into their device to trigger a cryptographic signature using a private key stored inside a tamper-proof hardware secure element.
3. Neutralizing AI Social Engineering: Because the hardware key only negotiates directly with the real web domain over an encrypted browser session, it cannot be intercepted. Even if an election employee is thoroughly deceived by an AI-cloned phone call, there is no text code or push button for them to read or pass to the attacker. The physical token cannot be socially engineered over a voice channel.

3.3 SIEM with User and Entity Behavior Analytics (UEBA)

A. Justification

Modern cyberattacks operate at a speed that human security teams are unable to manually monitor or thwart, especially when they are expedited by automated AI toolsets. Distributed Denial of Service (DDoS) network floods, which are intended to interfere with public updates and access to official portals during crucial tabulation windows, pose a serious threat to the agency, as noted in Section 1.3. The agency's centralized intelligence foundation is a Security Information and Event Management (SIEM) system supplemented with machine learning-based User and Entity Behavior Analytics (UEBA). In order to quickly identify complicated, fast-moving anomalies that conventional rule-based detection systems overlook, it collects data from all networks.

B. Function

The SIEM platform acts as a unified data lake, ingesting millions of log telemetry rows per second from perimeter Firewalls, Web Application Firewalls (WAF), ZTNA gateways, identity systems, and cloud databases.

1. Behavioral Baselineing: The UEBA component continuously tracks and establishes a historical "normal behavior" profile for every user account and technical asset (e.g., regular login times, standard data transfer volumes, and authorized API endpoints).

2. Anomalous Anomaly Detection: If an internal credential suddenly initiates an automated script attempting to download thousands of rows of voter registration data at 3:00 AM from an unexpected IP address, the UEBA algorithms immediately flag it as a severe behavioral anomaly.
3. Automated Incident Isolation: The SIEM correlates this anomaly with perimeter firewall metrics (such as a concurrent spike in web traffic) and orchestrates an automated response via SOAR (Security Orchestration, Automation, and Response) scripts instantly locking the compromised account and isolating the target server from the internet before data exfiltration can occur.

3.4 Media Provenance Framework (C2PA Integration) & Secure SIP Telephony

A. Justification

The core challenge highlighted in the Al Jazeera case study is the uncontrolled proliferation of deepfake media and fake political personae that exploit the agency's communication channels to erode national trust. Traditional defenses cannot distinguish between a real audio/video announcement issued by the Election Commission and an AI-generated clone distributed by a foreign adversary. To protect the Official Communications & Media Assets detailed in Section 1.2, the agency must actively establish an unforgeable, verifiable stamp of authority on all digital output and securely validate inbound telecommunications channels.

B. Function

The agency deploys a dual-pronged media verification and secure routing system:

1. C2PA Content Credentials: The agency implements the Coalition for Content Provenance and Authenticity (C2PA) open technical standard across its media production workflows. When the agency records an official public statement or election update, the recording software uses asymmetric cryptography to bind a signed, tamper-evident manifest metadata block directly into the file container (e.g., JPEG, MP4) at the point of creation. This manifest is cryptographically signed using an X.509 digital certificate from a trusted Certificate Authority.
2. Public Verification: Major social media and news distribution networks read this manifest, displaying a "Content Credentials" verified badge to the public. If a malicious actor pulls the file down and attempts to alter a single pixel or audio frame via AI generation tools, the cryptographic SHA-256 hash validation instantly breaks, rendering the badge invalid and alerting the public to the fake media.

3. STIR/SHAKEN Framework for Telephony Security: To directly eliminate the vulnerability used in the New Hampshire incident where threat actors used bulk Voice over IP (VoIP) and SIP (Session Initiation Protocol) trunking networks to broadcast caller-ID spoofed robocalls—the agency partners with top-tier telecom carriers to strictly enforce STIR/SHAKEN (Secure Telephone Identity Revisited / Signature-based Handling of Asserted information using toKENs) protocols. Every inbound and outbound call routed through agency infrastructure is cryptographically verified and given an "A-level attestation" certificate, validating that the caller identity is legitimate and fully authorized, preventing adversaries from spoofing official government helplines.

4.0 EVALUATION

4.0.1 Threat-Technology Alignment

The table below summarizes why each security technology was selected over its conventional alternative, directly mapping to attack stages identified in Part A.

Table 1. Justification Summary

Technology	Selected	Rejected Alternative	Key Reason for Selection
Network access	Zero Trust Architecture (ZTA) via ZTNA Gateways	Traditional VPN	VPN provides unrestricted network access after authentication. ZTA ensures ongoing per-session, per-resource verification — which prevents lateral movement even if credentials are stolen via AI social engineering.
Authentication	FIDO2 / WebAuthn Hardware Keys	SMS OTP / Authenticator Push Apps	SMS OTP and push notifications still depend on human decision-making — thus susceptible to AI voice impersonation of phone calls. FIDO2 binds cryptographically to the actual domain; there is no code to intercept or manipulate.
Threat detection	SIEM with UEBA	Standalone SIEM or manual log review	SMS OTP and push notifications still depend

			on human decision-making — thus susceptible to AI voice impersonation of phone calls. FIDO2 binds cryptographically to the actual domain; there is no code to intercept or manipulate.
Media authenticity	C2PA + STIR/SHAKEN	AI deepfake detection models	AI-driven detection models depend on probability prediction and may be defeated by next-gen deepfakes. C2PA ensures cryptographic proof of origin – altering any bit will make the SHA-256 hash invalid.

4.1 Zero Trust Architecture (ZTA) via ZTNA Gateways

A. Why ZTA — and Why Not a VPN?

In the 2024 election case, lateral movement was a critical attack stage — once an election worker's identity was compromised through AI voice cloning or phishing, attackers could move across internal systems toward electoral tabulation records. The classic VPN architecture gives a lot of access rights within the network after only one successful login process, which makes it useless once the attacker gains access using their own legitimate identity.

ZTA eliminates implicit trust entirely. Even if an attacker hijacks a valid employee account, they are restricted to only that one authorized application for that session and cannot move laterally into tabulation or voter registration systems. For a national election management agency — where a breach of tabulation records carries consequences beyond financial loss and directly threatens public trust in democracy — this structural containment justifies the higher implementation cost and operational complexity over conventional VPN.

B. Cost, Benefit & Feasibility

For enterprise ZTNA solutions such as Zscaler Private Access and Microsoft Entra Private Access, the cost of implementation per year amounts to approximately USD 150,000-500,000 for a mid-size government organization with approximately 500 employees. A study conducted by Forrester Consulting

in 2022 on the Total Economic Impact of Zero Trust revealed that organizations that adopted Zero Trust reduced their chances of success in a data breach significantly.

The most critical benefit in this context is micro-segmentation: internal networks are invisible to users and endpoints unless continuously re-verified per session. This directly neutralizes Part A Section 5.3 (lateral movement), where attackers attempt to move from compromised communication endpoints toward internal election systems.

C. Impact on Users

Election officers will access systems through a ZTNA portal rather than a traditional VPN, requiring approximately two to four weeks of adaptation. Login time increases slightly due to device posture checks, though modern ZTNA platforms minimize this delay. Field officers must use pre-registered, compliant devices; an emergency access approval process should be established for exceptional circumstances. Long-term, the risk of organization-wide disruption from a single compromised account is significantly reduced.

D. Implementation Challenges

Legacy internal systems using LDAP-based authentication may require additional integration layers or phased migration. Policy engineering is complex — each user role, department, and resource must be mapped, which may take several months for a large government agency. Vendor dependence is a matter of worry, but following open standards like the NIST SP 800-207 can help mitigate risks of vendor lock-in. Well-planned change management and cybersecurity awareness programs are crucial to overcoming any resistance from employees.

4.2 Phishing-Resistant IAM via FIDO2 / WebAuthn

A. Why FIDO2 — and Why Not SMS OTP or Authenticator Apps?

Part A Section 3.1 demonstrated that generative AI tools can produce highly realistic voice clones for a trivial cost. An attacker can impersonate a senior election director to pressure a technical officer into reading aloud an SMS OTP or approving a push notification — a technique known as real-time phishing or MFA fatigue. Both SMS OTPs and authenticator push notifications remain exploitable because they depend on human judgment under pressure.

FIDO2 removes human decision-making from authentication entirely. The hardware key uses asymmetric cryptography, binding the credential directly to the legitimate election agency domain. Even if an employee is fully convinced by an AI-cloned voice call, there is no code to read out — the physical token cannot be socially engineered over a voice channel, making this the only authentication method truly resistant to AI-driven identity attacks in this context.

B. Cost, Benefit & Feasibility

Security devices like YubiKey 5 Series would cost between USD 50-85, resulting in USD 25,000-42,500 in total costs for a 500-user organization. IAM platforms (examples include Microsoft Entra ID P2 and Okta Workforce Identity) would cost USD 9-22 monthly per user. Total first-year implementation cost for 500 users is approximately USD 79,000–174,500.

The security benefit is substantial. A FIDO Alliance case study found that after Google deployed FIDO-based authentication across 85,000 employees, the organization experienced zero successful phishing attacks against employee accounts over multiple years. Authentication speed also improves — users touch a hardware key rather than waiting for OTP delivery.

C. Impact on Users

Users must carry a hardware security key at all times. Organizations must establish secure issuance, revocation, and backup key procedures for critical personnel during active election periods. Temporary and field workers may require structured onboarding due to limited familiarity with hardware authentication. The primary long-term benefit is reduced psychological pressure — employees no longer need to judge whether a phone call or request is legitimate, as the hardware key cannot be deceived.

D. Implementation Challenges

Hardware key loss or damage requires a managed recovery process that is itself strongly verified, to prevent attackers from exploiting account recovery as an alternative attack pathway. Legacy election applications may require middleware integration before supporting FIDO2/WebAuthn standards. Supply chain security must be enforced — all hardware tokens must be procured from verified, trusted vendors to eliminate the risk of compromised devices entering government infrastructure.

4.3 SIEM with User and Entity Behavior Analytics (UEBA)

A. Why SIEM/UEBA — and Why It Cannot Stand Alone?

Part A Section 2.4 identified that attackers used AI and big data analytics to collect voter behavior and interaction data at scale, while Part A Section 4.3 highlighted that organizations typically respond only after misinformation or breaches have already spread. SIEM with UEBA addresses this reactive gap by continuously correlating security events across all systems in real time.

SIEM alone matches known attack signatures, but modern AI-accelerated attacks may not match any predefined rule. The UEBA layer adds machine learning-based behavioral baselining — detecting deviations from normal user and entity behavior even for novel, unseen attack patterns. Critically, SIEM/UEBA functions as the centralized intelligence layer that connects ZTA, FIDO2, and cloud systems: a compromised account attempting bulk voter data downloads at an unusual hour may trigger separate low-severity events in three different systems, only appearing as a critical incident when correlated by SIEM.

B. Cost, Benefit & Feasibility

The enterprise SIEM/UEBA solutions (Microsoft Sentinel, IBM QRadar, Splunk Enterprise Security) would have a subscription cost ranging between USD 80,000-250,000 annually, based on log size and level of automation, together with an additional recruitment of SOC staff members, from 3 to 5 cybersecurity experts.

According to the report by IBM Security "Cost of a Data Breach" for 2023, firms utilizing AI-enabled cybersecurity solutions and automation had a lifecycle of data breaches more than 100 days shorter compared to companies without such tools. For 2024, the same company discovered that faster detection and automation (through SOAR) greatly decreased the duration and expenses associated with a data breach. In election environments — where any delay during voting or tabulation could allow data manipulation before discovery — these capabilities are considered essential rather than optional.

C. Impact on Users

General election staff experience minimal workflow disruption because SIEM operates as a background monitoring system. SOC analysts take on a more active role managing alerts and incident response workflows. Automated SOAR actions may occasionally block legitimate users during unusual-hours

access (e.g. emergency audits), requiring clear escalation and unblock procedures. Alert thresholds must be carefully tuned to handle elevated system activity during active election periods without overwhelming the SOC team.

D. Implementation Challenges

Alert fatigue from false positives is the primary operational risk — UEBA behavioral baselines require months of accurate data collection before reliable anomaly detection is achieved. Infrastructure scalability must be validated, as SIEM requires sufficient processing and storage capacity to ingest logs from all perimeter and internal systems without gaps. Recruiting qualified cybersecurity professionals for SOC operations is challenging in the public sector due to salary competition with private industry. Behavioral monitoring policies must be reviewed for compliance with labor regulations and communicated transparently to staff.

4.4 Media Provenance Framework (C2PA) & Secure SIP Telephony (STIR/SHAKEN)

A. Why This Technology — and Why It Is Unique to This Organisation Type?

While ZTA, FIDO2, and SIEM are applicable across many organizations with sensitive digital infrastructure, C2PA and STIR/SHAKEN are specifically chosen because the election agency functions as a public communication authority. Part A Section 4.1 identified that no reliable global system currently verifies whether digital media content is authentic — a limitation that, for most organizations, creates a public relations issue, but for an election agency creates a direct attack surface where adversaries can impersonate official government announcements using deepfake technology.

Other technologies in the proposed architecture address identity protection, network access, or threat monitoring — none verify the authenticity of outbound media content or inbound telephony. C2PA and STIR/SHAKEN close this gap directly. Evidence from practice: In September 2024, the FCC levied a fine on Lingo Telecom in the amount of USD 1 million for breaking STIR/SHAKEN regulations in relation to the robocalls created using AI and imitating President Biden that targeted voters in New Hampshire.

B. Cost, Benefit & Feasibility

C2PA integration costs approximately USD 30,000–80,000 as a one-time engineering cost, with ongoing expenses limited to annual X.509 certificate renewal. STIR/SHAKEN introduces minimal additional cost as implementation is already required at the telecommunications carrier level under FCC regulations.

C2PA provides cryptographic proof-of-origin: any AI-generated modification to a signed media file — including a single pixel or audio frame — immediately invalidates the SHA-256 hash signature, rendering the verification badge invalid. Unlike deepfake detection models that rely on probabilistic AI prediction, cryptographic verification offers a mathematically deterministic method for confirming media authenticity, making it significantly harder to evade.

C. Impact on Users

Media production staff experience minimal workflow disruption — C2PA signing is integrated into existing content creation tools and triggers automatically on export. The general public benefits through visible Content Credentials verification badges on supported platforms, reducing susceptibility to deepfake impersonation. STIR/SHAKEN ensures outbound agency calls carry cryptographically verified caller identity, improving recognition by citizens. A key limitation is platform adoption dependency: verification badges are only visible on platforms that support C2PA manifest reading.

D. Implementation Challenges

C2PA verification effectiveness depends on external platform support — the agency can sign all official content, but verification is only visible where the receiving platform implements C2PA manifest reading. Certificate management requires ongoing maintenance; expired or compromised X.509 certificates could invalidate verification for legitimate official content and cause public confusion. C2PA cannot protect previously published unsigned media retroactively. STIR/SHAKEN coverage is incomplete for telecommunications traffic passing through older non-IP telephone infrastructure, meaning not all spoofed robocalls can be prevented across every channel.

4.5 Evaluation Summary

The table below consolidates cost, benefit, and primary implementation challenge for all four recommended security technologies.

Table 2. Evaluation summary

Technology	Estimated Cost (500 Users)	Key Benefit	Primary Challenge
ZTA via ZTNA	USD 150,000–500,000 / year	Eliminates lateral movement; restricts access to only authorized applications per session	Legacy LDAP system compatibility; weeks of policy engineering required

FIDO2 / WebAuthn	USD 79,000–174,500 first year (hardware + IAM platform)	Zero successful phishing attacks post-deployment (Google, 85,000 employees, FIDO Alliance)	Hardware key management; supply chain risk; temporary worker onboarding
SIEM with UEBA	USD 80,000–250,000 / year + SOC staff (3–5 FTEs)	100+ day reduction in breach lifecycle (IBM 2023/2024); automated SOAR isolation	Alert fatigue if baselines misconfigured; talent acquisition cost
C2PA + STIR/SHAKEN	USD 30,000–80,000 one-time + annual cert renewal (STIR/SHAKEN at carrier level)	Only cryptographic media verification — any AI edit breaks SHA-256 hash; FCC-enforced telephony	Platform adoption dependency; no retroactive signing of existing media

PART C

Reflection

The most challenging part of this assignment was understanding how AI-driven misinformation attacks differ from traditional cybersecurity incidents. Unlike conventional attacks that target software vulnerabilities or network systems, the 2024 U.S. election deepfake incident exploited human cognition, public trust, and social media algorithms — making it difficult to address through standard technical controls alone.

Initially, our group approached the solution design by identifying common technologies such as firewalls, SIEM, and IAM as general security controls. However, we later realized that simply listing technologies was insufficient. The real challenge was mapping specific attack activities to organizational systems and identifying where security controls could realistically reduce risk. For example, AI voice cloning targets authentication, compromised identities enable lateral movement toward tabulation systems, and deepfake media impersonates official government announcements — each requiring a different technical countermeasure.

We also recognized that some threats, such as external deepfake generation and social media bot amplification, cannot be fully prevented through internal network security architecture alone. Rather than viewing this as a weakness, we understood that acknowledging architectural limitations is part of strong cybersecurity analysis — a realistic design must identify which threats can be mitigated, which can only be reduced, and which require broader cooperation such as policy enforcement and platform regulation.

This assignment taught us that cybersecurity analysis requires threat-driven thinking, evidence-based justification, and context-specific solutions — skills that are increasingly essential as the threat landscape continues to be reshaped by artificial intelligence.

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